

2 Algoritmos probabilistas

1. Let $A[1 \dots n]$ be an array with n different integers. Let $ran(n)$ a randomized function that outputs an integer i , $1 \leq i \leq n$ under the uniform distribution. Let us consider the following algorithms

X **Input:** $A[1 \dots n]$
from $i = 1$ **to** n **do**
 swap values $A[i]$ and $A[ran(n)]$

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crea problemaz pe la si ajutoz!

2. Consider the set $S = \{1, \dots, n\}$.

$$\frac{1}{2^n}$$

- (a) We generate $X \subseteq S$ as follows: A fair coin is flipped independently for each element of S , if the coin lands H, the element is added to X , otherwise it is not. Prove that the resulting set X is equally likely to be any one of the 2^n possible subsets.
- (b) Suppose $X, Y \subseteq S$ are chosen independently and u.a.r. from all 2^n subsets from S . Compute $\Pr[X \subseteq Y]$ and $\Pr[X \cup Y = S]$

$$P[X \subseteq Y] = \sum_{i=0}^n P_n(|Y|=i) \cdot P_n(X \subseteq Y \mid |Y|=i)$$

$\uparrow$ de la mantin 0 in X
 $\downarrow$ $\downarrow$
 $\frac{\binom{n}{i}}{2^n}$ $\frac{1}{2^{n-i}}$

$$P[X \subseteq Y] = \left(\frac{3}{4}\right)^n$$

$$P[X \cup Y = S] = \left(\frac{3}{4}\right)^n \quad \text{e el mateix} \quad P[X \subseteq Y] \Leftrightarrow$$

3. Consider the following algorithm to generate an integer $r \in \{1, \dots, n\}$: We have n coins labelled $m_1, \dots, m_n$, where the probability that $m_i = \text{head}$ is $1/i$. Toss the coins in order until getting the first head, if the first head appears with coin m_i , the $r = i$. Prove that the previous algorithm yields an integer r with uniform distribution. i.e. the probability of getting any integer r is $1/n$.

$$\begin{array}{ccccccc}
 m_n & m_{n-1} & m_{i+1} & m_i & & & \\
 \text{+} & \text{+} & \dots & \text{+} & \text{h} & \text{ } & \\
 1 - \frac{1}{n} & 1 - \frac{1}{n-1} & 1 - \frac{1}{i+1} & \frac{1}{i} & & & \\
 & & & & (i+1) & (i) &
 \end{array}
 \quad
 1 - \frac{1}{(i+1)} \equiv \frac{(i+1)-1}{i+1}$$

$$P(i) = \left(1 - \frac{1}{n}\right) \cdot \left(1 - \frac{1}{n-1}\right) \cdot \dots \cdot \left(1 - \frac{1}{i+1}\right) \cdot \left(\frac{1}{i}\right)$$

$\Downarrow$

$$P(i) = \left(\frac{n-1}{n}\right) \cdot \left(\frac{n-2}{n-1}\right) \cdot \dots \cdot \left(\frac{i}{i+1}\right) \cdot \left(\frac{1}{i}\right)$$

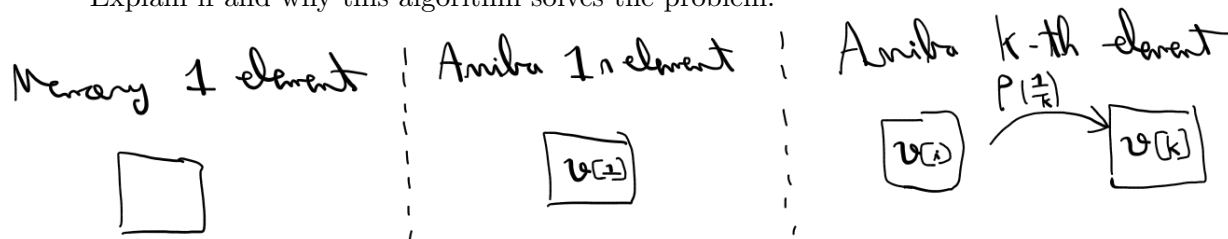
$$\left[P(i) = \underbrace{\left(\frac{i}{n}\right)}_{\text{tail}} \cdot \underbrace{\left(\frac{1}{i}\right)}_{\text{head}} = \frac{1}{n} \right]$$

DIC QUE SÓN EVENTS
INDEPENDENTS!

4. The following approach is called **reservoir sampling**. Suppose we have a sequence of items passing by one at a time. We want to maintain a sample of one item with the property that it is uniformly distributed over all the items that we have already seen at each time step. Moreover, we want to accomplish this without knowing the total number of items in advance or storing all of the items that we see.

Consider the following algorithm, which stores just one item in memory at all times. When the first item appears, it is stored in the memory. When the k -th item appears, it replaces the item in memory with probability $1/k$.

Explain if and why this algorithm solves the problem.



$$P(k) = \begin{matrix} & 1 & 2 & 3 & \dots & k-1 & k \\ \begin{matrix} 1 \\ \frac{1}{1} \end{matrix} & \begin{matrix} 2 \\ \frac{1}{2} \end{matrix} & \begin{matrix} 3 \\ \frac{1}{3} \end{matrix} & \dots & \begin{matrix} k-1 \\ \frac{1}{k-1} \end{matrix} & \begin{matrix} k \\ \frac{1}{k} \end{matrix} \end{matrix}$$

per me un element
un random()

• Prob de que entri element al moment i a rezultat al k

$$P(k) = \left(\frac{1}{i}\right) \cdot \left(1 - \frac{1}{i+1}\right) \cdot \left(1 - \frac{1}{i+2}\right) \cdot \dots \cdot \left(1 - \frac{1}{k-2}\right) \cdot \left(1 - \frac{1}{k-1}\right) \cdot \left(1 - \frac{1}{k}\right)$$

$$P(k) = \left(\frac{1}{i}\right) \cdot \left(\frac{i}{i+1}\right) \cdot \left(\frac{i+1}{i+2}\right) \cdot \dots \cdot \left(\frac{k-3}{k-2}\right) \cdot \left(\frac{k-2}{k-1}\right) \cdot \left(\frac{k-1}{k}\right)$$

$$P(k) = \left(\frac{1}{\cancel{i}}\right) \cdot \left(\frac{\cancel{i}}{\cancel{i+1}}\right) \cdot \left(\frac{\cancel{i+1}}{\cancel{i+2}}\right) \cdot \dots \cdot \left(\frac{\cancel{k-3}}{\cancel{k-2}}\right) \cdot \left(\frac{\cancel{k-2}}{\cancel{k-1}}\right) \cdot \left(\frac{\cancel{k-1}}{k}\right)$$

$$\left[P(k) = \left(\frac{1}{1}\right) \cdot \left(\frac{1}{k}\right) = \left(\frac{1}{k}\right) \right]$$

5. We have a function $F : \{0, \dots, n-1\} \rightarrow \{0, \dots, m-1\}$. We know that, for $0 \leq x, y \leq n-1$, $F((x+y) \bmod n) = (F(x) + F(y)) \bmod m$. The only way we have for evaluating F is to use a lookup table that stores the values of F . Unfortunately, an Evil Adversary has changed the value of $1/5$ of the table entries when we were not looking.

Describe a simple randomised algorithm that, given an input z , outputs a value that equals $F(z)$ with probability at least $1/2$. Your algorithm should work for every value of z , regardless of what values the Adversary changed. Your algorithm should use as few lookups and as little computation as possible.